CHL5233: Quarto in-class assignment

Qiulei Xu

For this in-class assignment, you will work in a group to create your **individual** Quarto document. Please consult your classmates and Tutorial: Authoring webpage by Quarto. You can also use the Internet including ChatGPT. The TA and I are also available for any questions.

When you are finished, render the document to PDF. Please carefully **proof read** and make edits as needed. **Commit** and **push** the edits to this file and the new PDF file to your **new repository**.

Create a new repository

- Under your GitHub account, create a **new** respository called `armed_conflict`
  - Throughout the course, we will use this repository to **reproduce** and **replicate** the analysis in the paper
- Create **two** folders
  - `data`
  - `reports`
- Save `countries.txt` and `regions.txt` into `data/raw/` subfolder
- Save `quarto_inclass.qmd` into `reports/` folder
- Eventually, your repository will have a very organized structure like this:

Example of data science organization

```
armed_conflict/
├── README.md
├── data/
│   ├── raw/
│   └── processed/
├── scripts/
│   ├── 01_import.R
│   ├── 02_clean.R
│   └── 03_analysis.R
├── outputs/
│   ├── figures/
│   └── tables/
└── reports/
    └── analysis.qmd
```

Edit this file to create a PDF document using Quarto

The R code chunk below includes the R packages required to knit this document. This is often called a **setup** code. We want the code to run, but we don't want the final document to show the

code or results. Edit the code chunk so that the code will run but the code and results (including all warnings and messages) are hidden from the final PDF output.

```
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.3
```

```
Warning: package 'ggplot2' was built under R version 4.5.3
```

```
Warning: package 'tibble' was built under R version 4.5.3
```

```
Warning: package 'tidyr' was built under R version 4.5.3
```

```
Warning: package 'readr' was built under R version 4.5.3
```

```
Warning: package 'purrr' was built under R version 4.5.3
```

```
Warning: package 'dplyr' was built under R version 4.5.3
```

```
Warning: package 'stringr' was built under R version 4.5.3
```

```
Warning: package 'forcats' was built under R version 4.5.3
```

```
Warning: package 'lubridate' was built under R version 4.5.3
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(knitr)
```

```
Warning: package 'knitr' was built under R version 4.5.3
```

```
library(kableExtra)
```

```
Warning: package 'kableExtra' was built under R version 4.5.3
```

```

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

    group_rows
```

For this exercise, we will use the following two data sets.

1. `countries.txt`: A data set that contains a list of countries used in the paper "Implications of armed conflict for maternal and child health: A regression analysis of data from 181 countries for 2000–2019" and their three-digit ISO (International Organization for Standardization) code.
2. `regions.txt`: A data set that contains a list of all major countries in the world and their
    a. ISO code (`country_code`)
    b. region (`region`)
    c. sub-region (`sub_region`)
    d. numerical region code (`region_code`)
    e. numerical sub-region code (`sub_region_code`)

The **final goal** is to display a **bar chart** that depicts how many countries are in each region.

Read in data and create an analytical data set

First, edit the code chunk below that reads in `countries.txt` to also display the number of rows and columns of the data.

```
countries <- read.table("../data/raw/countries.txt", header = TRUE)
nrow(countries)
```

```
[1] 186
```

```
ncol(countries)
```

```
[1] 2
```

There are 186 countries included in the data set.

Second, read in `regions.txt` and also write a sentence that states how many countries are included in the data set.

```
regions <- read.table("../data/raw/regions.txt", header = TRUE)
nrow(regions)
```

```
[1] 249
```

```
ncol(regions)
```

```
[1] 6
```

There are 249 countries included in the data set.

Third, subset the `regions` data set to only include countries from the `countries` data set.

```
sub_regions <- regions[regions$country_code %in% countries$ISO, ]
nrow(sub_regions)
```

```
[1] 186
```

```
nrow(countries)
```

```
[1] 186
```

Tabulate countries by region and sub-region

Next, we want to find out how many countries are in a region. Use the `group_by()` and `summarise()` functions in the `dplyr` package to count the number of countries in each region. Then use the `kable()` function in the `knitr` package to create a simple table with the column headers, "Region" and "Count".

```
regions |>
  group_by(region) |>
  summarise(Count = n()) |>
  kable(col.names = c("Region", "Count"))
```

Region	Count
	1
Africa	60
Americas	57
Asia	51
Europe	51
Oceania	29

Create a similar table that shows the number of countries in each **sub-region**.

```
sub_regions |>
  group_by(region) |>
  summarise(Count = n()) |>
  kable(col.names = c("Region", "Count"))
```

Region	Count
Africa	54
Americas	35
Asia	46
Europe	40
Oceania	11

Data visualization using ggplot

Here, we will use `geom_bar()` function within `ggplot()` to create a bar chart that displays the **number of countries** in each **sub-region**. 1. Color code each sub-region by **region**. 2. Create bars that run horizontally where sub-region names appear along the y-axis. 3. Order the bars such that the sub-region with the most number countries appear at the top and the sub-region with the least number of countries appear at the bottom.

```
ggplot(
  sub_regions,
  aes(
    y = fct_rev(fct_infreq(sub_region)),
    fill = region
  )
) +
  geom_bar() +
  labs(
    title = "Number of Countries by Sub-region",
```

```
    x = "Number of Countries",
    y = "Sub-region",
    fill = "Region"
  )
```

